SENTHAMILAN A

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PROFESSIONAL SUMMARY

Results-driven Data Analyst with 1 year of hands-on experience analyzing high-volume transactional datasets and building data-driven solutions using SQL, Power BI, and Excel. Experienced in transforming raw data into actionable insights through dashboard development, statistical analysis, and business performance reporting.

SKILLS

- **Data Visualization:** Power BI (DAX, custom visuals, bookmarks, KPI cards), Excel (PivotTables, Charts, Power Pivot)
- **Data Analysis & Querying:** SQL (Joins, Subqueries, CTEs, Window Functions, Aggregations), Python (pandas, NumPy — basics)
- **Analytics Techniques:** Trend Analysis, Anomaly Detection, Ad-Hoc Analysis, Reconciliation Reporting, Time-Series Analysis
- **Data Transformation:** Excel formulas (XLOOKUP, INDEX-MATCH, IFS), Power Query, SQL scripts, Data Modeling (Star Schema)
- **Soft Skills:** Problem Solving, Analytical Thinking, Business Communication

PROFESSIONAL EXPERIENCE

Data Analyst – Transaction Analytics

HectaData Pvt Ltd | Client: FSS (Canara Bank)| Jan 2024 - Jan 2025|

- Analyzed high-volume ATM, POS, and digital payment transaction data using SQL and Excel to uncover **decline patterns, monitor channel success rates, and surface anomalies** impacting transaction performance.
- Segmented transaction failure data by **error codes and issuer/acquirer channels** to identify recurring patterns and likely contributing factors behind declines.
- Built interactive Power BI dashboards tracking **channel-wise performance trends**, enabling faster identification of high-**failure-rate segments across the network.**
- Conducted historical trend analysis using Excel (Power Query, Pivot Tables) to benchmark **current decline volumes** against **past patterns** and highlight statistically significant **outliers** for review.
- Applied time-series analysis using SQL window functions **(LAG/LEAD)** to identify lag-effect correlations between **peak-hour traffic** surges and subsequent **timeout-related failure spikes.**

ANALYTICS EXPERIENCE

Payment Channel Performance & Failure Analytics | [Power BI, SQL, DAX, Power Query, Data Modeling]

Video Presentation • Live BI Dashboard • GitHub Repo

- **Problem:** No visibility into which payment **channels**, geographies, and time windows were **driving** the **highest transaction failure** rates across a banking payment network.
- **Action:** Built an 8-page Power BI dashboard analysing **598,496 banking transactions** modelled on **real payment domain patterns** from **professional experience**. Applied statistical anomaly detection **(Mean + 2SD)**, time-series spike analysis, and geographic risk segmentation across ATM, Online, and POS channels.
- **Result:** Identified **ATM** as **highest-risk channel at 21.12% failure rate**; detected peak-hour failure spikes of **24-25%** using baseline deviation analysis; flagged 24K statistically anomalous transactions exceeding **104.93-second** duration threshold; delivered 6 prioritized recommendations with supporting evidence.

PROJECTS

BI 360 – End-to-End Business Performance Dashboard | [Power BI, SQL, DAX, Power Query, Data Modeling]

Live BI Dashboard • LinkedIn • GitHub Repo

- **Problem:** No unified view of $3.74B sales to track regional profitability
- **Action:** Designed a multi-view Power BI dashboard using **DAX time intelligence and data modeling across Finance, Sales, Marketing, and Supply Chain** functions to track **regional P&L and forecast accuracy**.
- **Result:** Exposed **$217M loss in India** (**-23%** net profit) despite **$945M sales**; Identified **COGS** overrun **(62% of revenue)**,Flagged **7** high-risk **OOS markets** (e.g., **Australia FC Accuracy: 18.58%**), which can reduce stockouts by 30%

Telecom Sector Analysis | [Power BI,SQL,DAX]

Video Presentation • GitHub Repo • Live BI Dashboard

- **Problem:** No clear visibility into 5G launch performance across regions to identify which plans and cities were driving churn and revenue loss.
- **Action:** Analyzed 8-month telecom data (13 plans, 8 cities) using a Power BI dashboard to evaluate KPIs — TAU, TUsU, ARPU, and market share — and identified regional performance gaps driving churn and revenue loss.
- **Result:** Uncovered a 5.8% market share decline despite minimal revenue impact (-0.5%), pinpointing a 72% revenue drop in Plan 7 and 77% churn surge in Lucknow; delivered actionable recommendations via stakeholder presentation video.

CERTIFICATIONS

- Advanced Excel - **Codebasics** (2024) [Certificate]
- Power BI Data Analytics - **Codebasics** (2025) [Certificate]
- SQL Advanced For Data Professionals - **Codebasics** (2025) [Certificate]

EDUCATION

Government College Of Engineering - Salem | **Aug 2018 - Jul 2022**

Bachelor of Engineering - Mechanical Engineering | CGPA - 7.8 /10